

Examiner reports

Q1.

Students displayed excellent knowledge here with over 80% choosing the correct third option. The fourth option was chosen by most who did not choose the third.

Q2.

For many students part (a) was a straightforward application of a practised routine:

- $F = ma$ applied separately to each particle
- T eliminated from the two equations
- Use of $s = ut + \frac{1}{2}at^2$ to complete the solution.

A minority started with a single equation, $1.8g - 1.2g = ma$, which received zero marks. An equation separated from its underlying mechanical principle is hard to grasp, as evidenced by more of these students using $m = 1.8$ rather than $m = 3$.

Some students chose the wrong *suvat* equation and had to apply a second to find t . Premature or even final rounding sometimes lost the accuracy mark, where 3 significant figures were expected.

Students frequently approached the demand in part (b) in the wrong way. They should understand that what is required is an assumption that is necessary, sensible, specific and not already stated in the question. Thus saying that there was “no friction at the peg” or that “the string had negligible mass” scored zero marks. Similarly “no other forces act” is too general. A minority realised the necessity of sufficient length of string.

It is often better if the answer starts with an “I have assumed that” to ensure that the assumption goes on to be properly explained. For example “I have assumed that there is no air resistance” is better than just “air resistance” where the assumption is not clear.

Q3.

In part (a), most students recognised the effect of the air resistance, although some thought that Jason was behind because he was not applying enough driving force. Since “Less air resistance for Jason” and “More air resistance for Laura” were both correct answers, an unattributed “Less air resistance” did not explain sufficiently.

For part (b), most students correctly calculated the deceleration for Laura, but some had this as a positive acceleration. Some had an unexplained F in their equation, $F - 40 = 64a$, and made little progress. Others replaced this F with 64 or $64g$, getting a positive a .

It was possible to complete the question in a variety of ways:

- use $v^2 = u^2 + 2as$ and show $s < 40$ when $v = 0$, $a = -0.625$ and $u = 6.944$
- use $v^2 = u^2 + 2as$ with $a = -0.625$ and show that $v = 0$ at $s = 40$ as long as $u < 7.07$
- use $v^2 = u^2 + 2as$ with $v = 0$, $s = 40$ and $u = 6.944$ and deduce that $a < -0.603$
- as above but go back one stage to show resistive force must be > 38.5 N
- use $v^2 = u^2 + 2as$ with $a = -0.625$, $s = 40$ and $u = 6.944$ to get $v^2 = -1.8$ and deduce that Laura had come to rest before this.

The first method was the most popular, but clear solutions using all other methods were seen. All of these methods lost accuracy marks when 25 was used as the value of u .

For part (c), the assumptions needed to be necessary, sensible, specific and not already stated in the question. Since the total resistive force was given, this was one case where “air resistance” was not acceptable. A minority recognised that reaction time had been ignored, or that the resistive force might diminish as Laura slowed. Many stated that they had assumed there was no friction (between bike and road), when in fact friction here is essential for the brakes to work.

Q4.

Nearly all students made good progress with part (a), with over half scoring full marks. Nearly all students successfully found the maximum possible friction and made a comparison with the applied horizontal force. The most frequently seen error was when students wrote a statement that was incorrect, often along the lines of “friction is greater than 150N so there is equilibrium”. This showed a misunderstanding of the friction model and could not be overlooked.

Part (b) was also well answered, but a lack of detail often meant that students had not fully justified their answer. Several impressive and detailed answers were seen, where students explained clearly that the horizontal component of the applied force was greater than the maximum friction so that the crate would not remain stationary.

Resolving of forces was generally successful.

Q5.

A high proportion of students had great difficulty in forming an appropriate equation of motion for part (a)(i). Those who attempted to treat the system as a whole made mistakes such as using the wrong total mass, including incorrect tension forces, using weight instead of mass or forgetting to include the resistance on the buggy. Some students considered the skater and buggy separately and similar errors were seen, however a fairly common error was to subtract the mass of the skater from the mass of the buggy in the equation of motion of the buggy.

Those students who wrote down a correct equation of motion usually completed the question successfully. Approximately half of the students scored full marks on part (a)(i).

In part (a)(ii) students were still able to make progress even if they had gone wrong in (a)(i). Provided their equation of motion was correct using their R they were given credit. Over half of students scored at least 2 marks.

Part (b) was a request for assumptions in a very standard model. A significant number of students gave assumptions that did not need to be made, as they were given in the question. For example, it was unnecessary to assume the road was flat as the question stated “horizontal road”. Students should also avoid words like “flat” or “straight” when they mean “horizontal”. A correct assumption would be that the rope was horizontal.

In part (c)(i), students who had gone wrong in earlier parts of the question were still able to make good progress and they were credited for using correct techniques. Correctly using their incorrect R from part (a)(ii) often resulted in 4/5 marks. Common errors were caused by students over complicating the problem, often attempting, unnecessarily, to find the time taken for either the buggy or skater to stop.

Over half of students made some progress on (c)(ii), but the general standard of explanation was poor. A significant number of students showed a misunderstanding of the forces model, or may have been trying to over-explain. Statements like “the rope has been dropped so the buggy no longer feels the resistance from the skater” are not accurate. It is more concise and accurate to simply say “there is no tension acting on the buggy”.

Q12.

Again this question was done well by many students although some made minor errors. In part (c) a few students found the acceleration (as $1.4\mathbf{i} + 1.6\mathbf{j}$) and did not calculate the magnitude. Also, in part (d) some students found the angle as 41° rather than 49° . However many students gained full marks on this question.

Q13.

While this question was generally done well there were more errors than in the first two questions. In part (a) the most common error was to include a force directed down the slope. Part (b) was done well, but a few students gave the answer 29.4 and a few used $\sin 40^\circ$ instead of $\cos 40^\circ$. Part (c) was done very well and follow through marks were awarded to those with errors arising from part (b). Part (d) was a little more demanding, with the most common error being to work with only one force. There were many good answers to part (e), but some students gave answers that were not about the forces. For example, a statement like “The box is a particle.” did not relate to the forces acting on the box as was requested in the question.

Q14.

While there were many good solutions to this question, there were a number of errors in the solutions of some students. For example in part (a), some students ignored the tension in the tow bar acting on the tractor and created equations like $2500 - R = 3500 \times 0.2$. A number of students were able to find R correctly by finding the tension in the tow bar first before considering the forces on the tractor to find R . It was not uncommon to see students gain no marks in part (a), but gain full marks in part (b), when they considered the forces acting on the trailer correctly. Many students realised that the answer for part (c) was expected to be the same as the answer to part (b) and simply stated this as expected. However, some worked with the forces acting on the tractor to find the tension and produced an equation rather than simply stating the value.

Q15.

This question was usually answered well. However some students forgot the minus sign in

the initial equation $12 \frac{dv}{dt} = -4v^{\frac{1}{3}}$ whilst others had difficulty with the evaluation of $\int v^{-\frac{1}{3}} dv$.

Most students attempted to find the ‘+c’ term and obtained the printed result. Nearly all students correctly found the time when the particle came to rest.

Q16.

Part (a) was a very standard question and many students did well with this part of the question. However, a small number of students ignored the instruction to form two equations of motion. In part (b), some students did not convert the 40 cm to metres and

obtained speeds that related to moving 40 m rather than 40 cm. These students did gain some marks, but lost the final accuracy mark.

In part (c) a common error was to use the wrong acceleration, with some students continuing to use 4.9 m s^{-2} rather than -9.8 m s^{-2} . In this part there was also some confusion between the values to use for u and v , with some students taking u as zero at the maximum height.

Most students were able to say that the acceleration would be reduced, when answering part (b), but a much smaller proportion of the students could give a good explanation.

Those who mentioned the effect of the air resistance on the resultant force normally gained both marks.

Q17.

Part (a) was often done very well by students, with a reasonable number of students gaining all 11 marks. There were a few issues with the force diagram, with missing or additional forces. Many students seemed to have been helped by the answer that was given for the normal reaction and this in turn enabled many to find the friction correctly. Finding the acceleration was also done well, although there were a few sign errors and some students simply used 150 instead of $150 \cos 20^\circ$. Part (b) was much more demanding and there were far fewer correct responses. The most common error was not to take account of the changed angle when finding the normal reaction and to simply continue to use the value of 243 from part (a).

In part (c) there were some good responses, but a fair number of students incorrectly stated that the tension would be greater for a greater constant speed.

Q18.

Nearly all students answered this question well.

Q19.

This question was generally done well. There were very many good force diagrams, with the most common error probably being the omission of arrow heads. Part (b) was done very well. While there were many good answers to part (b) some candidates simply calculated $4 \times 3 = 12$. There were also a few candidates who made a sign error, producing equations such as $F - 50 = 12$. Candidates were generally able to apply $F = \mu R$ to their answers from the previous parts, and many gained the follow through marks that were available. There were many good explanations for part (c), but also a number of confused answers, in which it was claimed that the coefficient of friction would have to increase.

Q20.

Candidates tended to do either very well or very badly on this question, although some were able to gain follow through marks in part (b) from an incorrect answer to part (a). Those candidates who realised that they had to consider the whole system in part (a) and did not include tensions usually obtained a correct solution. A large range of errors were seen which included adding a tension force, considering the weight or having the wrong multiple of R for the total resistance. The candidates seemed to find part (b) easier and were often able to form an equation, which they solved by just considering the car or the caravan. It was most common for them to consider the caravan, and most realised that the tension was given by $T = 1600 + 2R$.

Q21.

There were very many good attempts at both parts of part (a) with a lot of correct answers. However, there was quite a bit of confusion with part (b). A very common wrong answer was 686 N, and it appears that there was some confusion between the words 'resultant' and 'reaction'. There were some good solutions to part (c), although this part of the question was more demanding. A considerable range of incorrect responses were seen, including the omission of g , as in $70 \sin \alpha = 58.8$, the use of $\cos$ instead of $\sin$ or the use of the normal reaction from part (b), as in $686 = 70g \sin \alpha$.

In the second part of part (c), it was not unusual to see the 30 N force with the wrong sign which produced the following incorrect equation: $70 g \sin \alpha = 58.8 - 30$. In part (d), there were many good answers where the candidates linked the resistance to the speed of the cyclist. However, some simply said things such as "the resistance will not be constant" and did not give a reason to support this answer.

Q22.

This question was also answered well. In part (a), a few candidates did not divide F by 50 accurately. In part (b), a number of candidates did not find their '+c' correctly, simply replacing their 'c' with $7\mathbf{i} - 4\mathbf{j}$. Others, on evaluating their 'c', took $-e^{-2t}$ when $t = 0$ to be +1 rather than -1.

Q23.

There were many good attempts at this question. In part (a)(i), many candidates were able to obtain the correct answer although some found the time and then the acceleration. There were only a few cases of candidates confusing u and v in their equations. In part (a)(ii), there were again many good responses, but in some cases inappropriate information was used, for example using a distance of 75m or a final speed of 10 m s^{-1} . In part (a)(iii) there were again many correct answers, but quite a few candidates did not find the magnitude of the force and gave negative answers. This was also the case in part (b), where candidates also sometimes added the 200 to their previous answer instead of subtracting it from it.

Q24.

The candidates made very little progress with this question and did not seem to be able to deal with the resolving required.

Q25.

There were many good attempts at this question. In part (a)(i), many candidates were able to obtain the correct answer although some found the time and then the acceleration. There were only a few cases of candidates confusing u and v in their equations. In part (a)(ii), there were again many good responses, but in some cases inappropriate information was used, for example using a distance of 75m or a final speed of 10 m s^{-1} . In part (a)(iii) there were again many correct answers, but quite a few candidates did not find the magnitude of the force and gave negative answers. This was also the case in part (b), where candidates also sometimes added the 200 to their previous answer instead of subtracting it from it.

Q26.

Part (a) was done well by some candidates, but there were a significant number who proceeded as if this was a two particles connected by a string over a peg type of problem. Many times a pair of equations such as $18g - T = 18a$ and $T - 12g = 12a$ were seen. This made it impossible for candidates to complete this part of the question.

In part (b)(i), some candidates were able to get back on track, as they had the correct equation for the 18 kg particle, but some did use the acceleration from part (a) instead of 3 ms^{-2} . In (b)(ii), there were many correct responses, but a few gave the answer 36 N, suggesting that there was some confusion between “reaction force” and “resultant force”. The responses to part (b)(iii) were quite mixed with some good solutions. It was not uncommon for candidates to simply assume that the friction was equal to 36 N. Other candidates tried to find the difference between a “tension” which was often incorrect and the 36 N.

There were a number of good answers to part (c), but some were too vague. For example just writing “particles” was not enough as it did not make the connection that it was the block that was being modelled as a particle.

Q27.

The force diagram asked for in part (a) was generally well drawn, but some candidates did lose marks for not including arrow heads or appropriate labels. The candidates who understood that they should resolve vertically to find the reaction force did well although some ended up with $R = 78.4 + 0.5T$ or equivalent. Others produced very confused responses.

Those who had a correct expression for the normal reaction often made some progress in part (c). There were quite a number of candidates who made promising starts but then made algebraic errors when solving for T . There were a range of errors that were made in setting up an equation of motion. These included only using the friction force, using T instead of $T \cos 30^\circ$ and omitting the ma term.

Q28.

This question was also answered well by virtually all candidates, who showed that they knew the techniques involved in answering it. In part (c) the majority of students integrated to find the position vector but a common error was to forget the $+c$ term and thus to ignore the initial position vector. Some found the $+c$ incorrectly assuming that c was zero without completing the necessary calculation.

Q29.

Most candidates answered part (a) correctly but some just wrote down the printed result.

In part (a) some candidates started with $m \frac{dv}{dt} = -9.8v$ but found that this did not give the printed result, and just invented a term in g . This was not accepted. Some were penalised for ignoring the m term or for changing signs.

In part (b) candidates were expected to use $\int \frac{1}{v-5} dv = -1.96dt$; some split $\int \frac{1}{v-5} dv$ into $\int \frac{1}{v} - \frac{1}{5} dv$. However there were many good answers to this part with both the integrations required being successfully completed. The difficulty which some candidates

had was in the simplification needed to convert $\ln(v - 5) = -1.96t + \ln 2$ into $v - 5 = 2e^{-1.96t}$ which often became $v - 5 = e^{-1.96t} + 2$

Q30.

This question was also done very well by the candidates. The main issues were in parts (a) and (b). In part (a), a number of candidates made errors when calculating the area under the graph. Often they would have one small error, which meant that their total was incorrect. In these cases, the candidates' distances were followed through into part (b). Both parts (b) and (c) were done well with very few candidates losing marks. In part (d), Newton's second law was applied in all but a few cases, but the issue was with the units of the mass. Some candidates converted to kg incorrectly, while others simply found the force as if the mass were 200 kg.

Q31.

Part (a), and in particular part (a)(i), caused problems for a number of candidates. The issue was that they did not consider the whole system or included a tension force in some inappropriate way. The candidates who found this difficult did not seem to appreciate the need to consider the whole system first. Some candidates did find the tension first and then considered the forces on the car to find P. Part (a)(ii) was done better than part (a)(i), but there was evidence of confusion as to which component of the system to consider and which forces were acting on it.

Part (b) was done very well and a high proportion of the candidates gained full marks on this part.

There were many good answers to part (c), but some candidates did make reference to the condition of the road surface and often included friction as if the car and caravan were sliding on the road surface.

Q32.

Some candidates produced very good clear solutions to this question. While there were many good force diagrams, some were confused and did not make use of arrows to indicate the forces. There were some issues with the direction of the tension and the 500 N force.

In part (b), the candidates who could start with a clear statement of the equation of motion, usually completed the question and gained full marks. For those who realised that an equation of motion was required, the most common errors were to omit a force or to take the wrong component. A few candidates omitted the ma term and treated the problem as if the forces were in equilibrium.

Q33.

Most candidates answered this question well. In part (a), a small number of candidates forgot to add, and find, the $+c$ term and thus ignored the initial position vector; a few candidates tried to use equations for constant acceleration. In part (c), some candidates made errors when finding the force by finding ma , which was $2(6\mathbf{i} - 8\mathbf{j})$, to be $12\mathbf{i} - 8\mathbf{j}$.

Q34.

Most candidates answered part (a)(i) correctly but some just wrote down the printed result.

In part (a)(ii), candidates needed to start from $F = ma$ and show the cancellation of the mass (65). Some were penalised for ignoring the m term or ignoring the minus sign.

In part (b), candidates were asked to show that $\int \frac{1}{v-2.45} dv = -\int 4 dt$ many simply used this equation and were penalised. There were, however, many good answers to this part, with both the integrations required being successfully completed. The difficulty which candidates had was in the simplification needed to convert $\ln(v-2.45) = -4t + \ln 17.15$ into

$v - 2.45 = 17.15e^{-4t}$, which often became $v - 2.45 = 17.15 + e^{-4t}$.

Q35.

This question was done well by many candidates. There were a few minor slips in part (a). Part (b) was a little more challenging. The most common errors were to omit one force, typically the weight leading to an answer of 160 N, or to make a sign error, for example $800g - T = 800 \times 0.2$ leading to a final answer of 7680 N.



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